

# PF32 SYNC Input Configuration

## +3.3 Volt Compatible SYNC Input

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# 1 Introduction

This document describes how to connect a pulsed light source to the PF32 Camera. The PF32 camera requires a pulsed light source for most applications. In the time-correlated single-photon counting mode, the photons are timed in reference to the light source. In order to synchronize the PF32 Camera with the light source a SYNC input is provided. This SYNC input must be connected to the light source electrical output or a light detector in order for the PF32 to function properly. Without the SYNC input connected, the measurement would be uncorrelated and therefore lose its meaning.

Your PF32 Camera SYNC input has been factory-configured to **+3.3 Volts**. It is essential that the pulsed light source electrical connection never exceeds the amplitude of +3.3 Volts. **Connecting a wrong signal to the SYNC input may damage the PF32 Camera. Consult Photon Force or your local distributor in case of any doubts.**

This document explains how to ensure that the correct signal is attached to the PF32 Camera.

## 1.1 Quick Connection Procedure

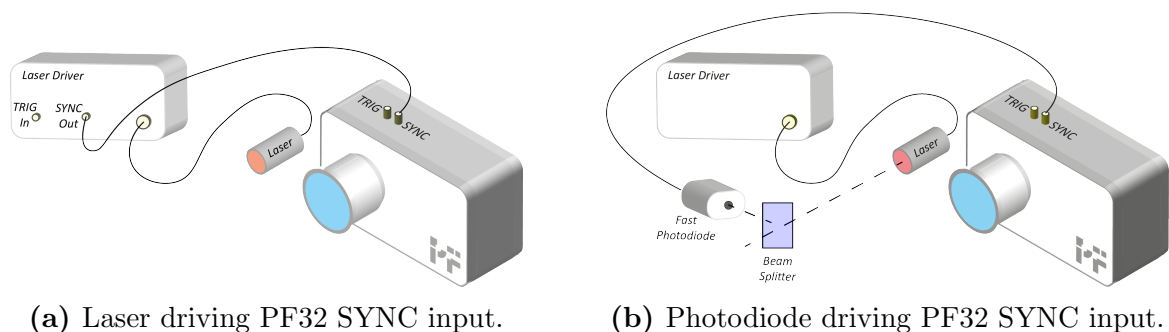
This document describes a safe connection procedure in detail. If you are using your PF32 Camera for the first time, familiarize yourself with the whole document. If you are familiar with the procedure, the following points provide guidance on the connection of the PF32 Camera to a pulsed light source:

- Check the SYNC input configuration of your camera.
- Connect the electrical output of the pulsed laser light source to an oscilloscope with  $50\Omega$  input impedance.
- Check that the voltage amplitude matches the nominal SYNC input voltage.
- Use an appropriate radio-frequency attenuator if the output voltage amplitude exceeds the nominal SYNC input voltage.
- Connect the PF32 Camera to the pulsed light source.



## 2 Pulsed Light Source Connection to PF32 Camera

The PF32 Camera is synchronized with optical pulses by an electrical connection to the light source or an external detector (**Figure 1**). The electrical pulses generated in the light source or fast optical detector are connected to the PF32 camera through the SYNC input. The SYNC input is through an SMA radio-frequency (RF) connector, one of seven on the PF32 (**Figure 1c**). Because the electrical properties of the trigger outputs vary for different pulsed light sources and light detectors, it is important to match them to the PF32 Camera prior to connection.



(c) PF32 Camera connectors. SYNC is the fifth gold SMA connector from the left.

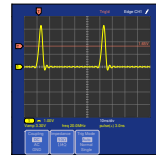
**Figure 1: Diagrams of the connection between a pulsed light source and the PF32 camera.** The connection between the light source and the PF32 Camera is facilitated by a coaxial cable. Schematic diagrams show a pulsed light source (a) or an external light detector (b) connected to the PF32 Camera. The generated electrical pulses synchronize the PF32 Camera with the optical pulses. (c) Image of the PF32 Camera showing the SYNC input as the fifth gold SMA connector from the left.

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### 3 Required Tools and Accessories

There is no common standard for outputs from pulsed light sources. It is therefore essential to understand the output trigger signal from your light source before it can be safely connected to the PF32 Camera. To assess the electrical properties of the light source trigger output, one or more of the following tools and accessories may be required:

- High-bandwidth oscilloscope



- SMA-ended RG-316 coaxial cable



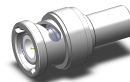
- 50  $\Omega$  SMA to BNC adapter



- 50  $\Omega$  BNC Tee adapter



- 50  $\Omega$  terminator



- 50  $\Omega$  feed-through terminator



- RF attenuator



Parts recommended by Photon Force are listed in **Appendix A** on page 14. Other parts with identical functionality may be substituted.

#### 3.1 Connecting the Oscilloscope

An oscilloscope is required to analyze the electrical output of the pulsed light source or optical pulse detector. The following paragraphs will only refer to a pulsed light source, but the same will apply for an optical pulse detector. The oscilloscope should have at least several hundred megahertz (MHz) bandwidth, although multi-gigahertz (GHz) bandwidth is advisable.

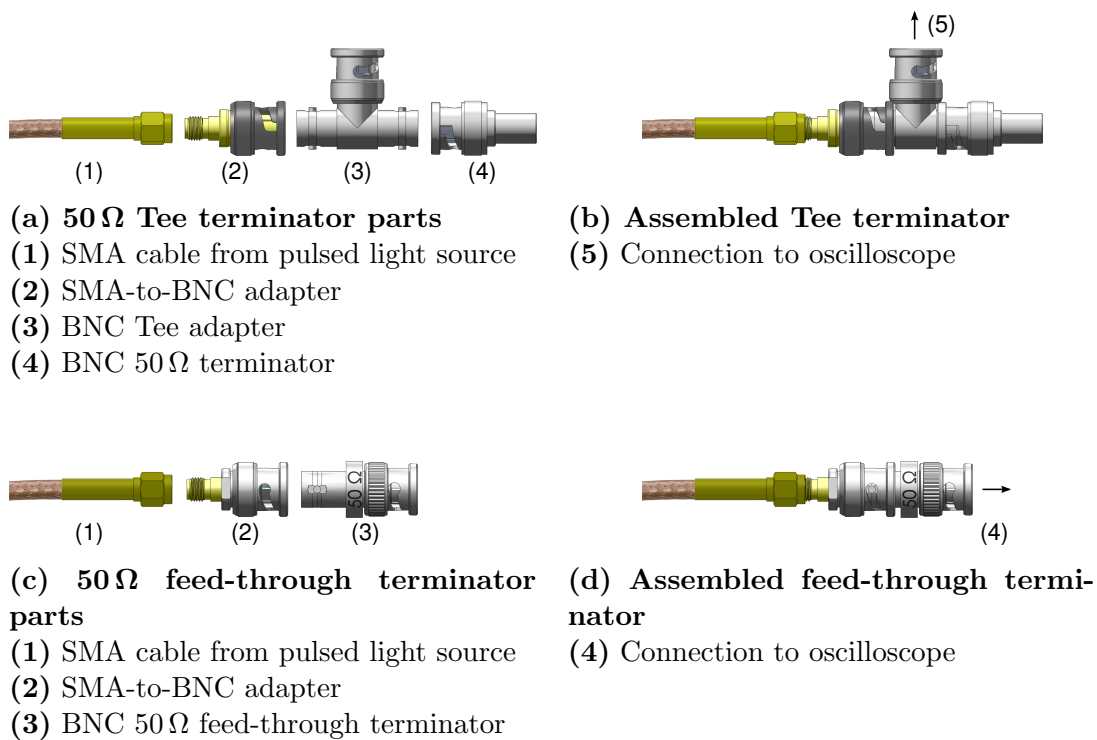
### 3.1.1 Input Impedance

The SYNC input of the PF32 Camera is terminated by a  $50\ \Omega$  resistor to ground. The coaxial cable connecting the pulsed light source therefore also needs to be terminated by a  $50\ \Omega$  load for the oscilloscope readings to correspond to the signal experienced by the camera. Many oscilloscopes support selection of their input impedance. Perform all measurements with  **$50\ \Omega$  oscilloscope input impedance**. For oscilloscopes that only offer a single option of  $1\ \text{M}\Omega$  input impedance, the cable must be terminated by a  $50\ \Omega$  load close to the oscilloscope's input. This can be achieved with a  $50\ \Omega$  BNC terminator and a  $50\ \Omega$  BNC Tee adapter or a feed-through  $50\ \Omega$  BNC terminator (**Figure 2**).

### 3.1.2 Connecting a Pulsed Light Source to the Oscilloscope

The PF32 Camera SYNC input is delivered through an SMA connector. The connection between the pulsed light source and the PF32 must therefore be done by a coaxial cable with an SMA connector at its end. Photon Force recommends using **RG-316 coaxial cables**. RG-316 coaxial cables offer high performance at an affordable price. Use a  **$50\ \Omega$  SMA to BNC adapter** to connect the SMA connector of the RG-316 coaxial cable to the oscilloscope. Use the same RG-316 coaxial cable that is intended to connect the PF32 Camera with the pulsed light source. Set the oscilloscope input impedance to  $50\ \Omega$  on the input connected to the pulsed light source. If the oscilloscope does not allow setting its input impedance to  $50\ \Omega$ , use a  $50\ \Omega$  BNC terminator with a  $50\ \Omega$  BNC Tee adapter or a feed-through  $50\ \Omega$  BNC terminator to connect the light source with the oscilloscope. See **Figure 2** explaining the connection of the pulsed light source to an oscilloscope without  $50\ \Omega$  input impedance support.





**Figure 2: External 50  $\Omega$  termination setup for oscilloscopes without a selectable input impedance.** Two setups are possible, either using a Tee-junction and an end terminator (**a-b, top row**), or a feed-through terminator (**c-d, bottom row**). The components for each terminator assembly are on the left (**a, c**). The assembled terminators and their connection to an oscilloscope input are on the right (**b, d**).

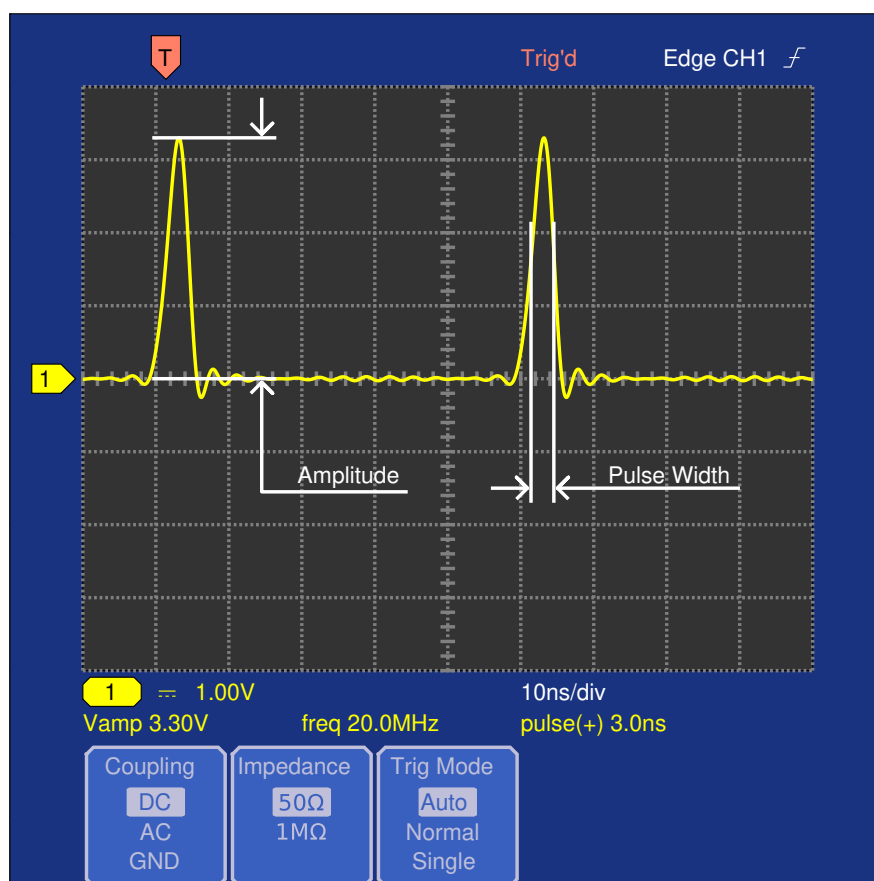
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## 4 Measuring Pulsed Light Source Electrical Output

Connect the electrical output of the pulsed light source to one of the oscilloscope's inputs as described in the above two paragraphs. Ensure that the end of the coaxial cable is loaded by a  $50\ \Omega$  termination provided either by the oscilloscope itself or an external terminator. Select **DC input coupling** on the channel with the connected pulsed light source. Set the oscilloscope to **Auto trigger mode**. Adjust the **trigger** level to match the input - set it to half the amplitude. Use the oscilloscope **Measure** function to assess the electrical properties of the pulse. Measure the **Voltage Amplitude** and **Pulse Width**.

An example screen of a correctly configured oscilloscope is in **Figure 3**. The screen shows a hypothetical pulsed light source connected through its electrical output to oscilloscope channel 1. The voltage amplitude is 3.30 V, the repetition frequency is 20 MHz and the pulse width is 3.0 ns. Voltage amplitude is the difference between the high and low voltage levels without the overshoots. Pulse width is measured at half of the amplitude.





**Figure 3: Typical oscilloscope screen showing electrical output of a pulsed light source.** To assess the electrical output of a pulsed light source, set the input coupling to DC, input impedance to  $50\Omega$  and trigger mode to Auto. The yellow curve is a typical output from a pulsed laser source. The measured characteristics of the electrical signal are voltage amplitude of 3.30 V, pulse repetition frequency of 20.0 MHz and pulse width of 3.0 ns.



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## 5 Adjusting Pulsed Light Source Output

The voltage amplitude of the pulsed light source electrical output should never exceed the SYNC input nominal voltage. Your camera SYNC input has been factory set to **+3.3 Volts**. The PF32 Camera will work with a lower voltage amplitude of pulsed light source electrical output. In fact, it is guaranteed to work down to  $0.7\times$  the SYNC input nominal voltage. However, the timing jitter will deteriorate as the voltage amplitude decreases. Photon Force always recommends matching the pulsed laser light source electrical output voltage amplitude to the SYNC input nominal voltage.

### 5.1 Attenuating Pulsed Laser Light Source Electrical Output

If the measured pulsed light source electrical output voltage amplitude exceeds the nominal SYNC input voltage, an attenuator must be inserted into the signal path. The choice of the attenuator depends on the difference between the pulsed light source output and the SYNC nominal voltage. The attenuators are specified in decibels (dB). The required attenuation ratio  $A$  is calculated from the voltage amplitude  $V_{light}$  of the pulsed light source electrical output and the nominal SYNC input voltage  $V_{SYNC}$  according to **Equation 1**.

$$A = 20 \cdot \log_{10} \left( \frac{V_{SYNC}}{V_{light}} \right) \quad (1)$$

For example, the output of a pulsed laser light source measured on an oscilloscope with  $50\ \Omega$  input impedance happens to be 3.6 Volts ( $V_{light}$ ). The PF32 Camera SYNC is configured for +3.3 Volts input ( $V_{SYNC}$ ). To choose the right attenuator, the above values are substituted into **Equation 1**. In this example the resulting value is  $-0.76\text{ dB}$ , according to **Equation 2**. Off-the-shelf attenuators are not available at arbitrary values and therefore an attenuator with a larger than calculated value should be used. In the above example, it would be an attenuator with a value of  $-1\text{ dB}$ . **Table 1** summarizes attenuators for various combinations of pulsed light source output voltages. It helps with the selection of the optimal attenuator for a given combination of pulsed light source and PF32 Camera SYNC input setting.

$$\begin{aligned} V_{SYNC} &= 3.3\text{ V} \\ V_{light} &= 3.6\text{ V} \\ A &= 20 \cdot \log_{10} \left( \frac{V_{SYNC}}{V_{light}} \right) = 20 \cdot \log_{10} \left( \frac{3.3\text{ V}}{3.6\text{ V}} \right) \text{ dB} \approx -0.76\text{ dB} \end{aligned} \quad (2)$$

The attenuators can be purchased from various suppliers. Photon Force recommends coaxial attenuators from the **FW** range made by **Mini-Circuits**. These attenua-



| <b>+3.3 V SYNC Input</b> |            |            |
|--------------------------|------------|------------|
| $V_{light}$              | Attenuator | $V_{SYNC}$ |
| 3.3 V                    | 0 dB       | 3.3 V      |
| 3.4 V                    | -1 dB      | 3.0 V      |
| 3.5 V                    | -1 dB      | 3.1 V      |
| 3.6 V                    | -1 dB      | 3.2 V      |
| 3.7 V                    | -1 dB      | 3.3 V      |
| 3.8 V                    | -2 dB      | 3.0 V      |
| 3.9 V                    | -2 dB      | 3.1 V      |
| 4.0 V                    | -2 dB      | 3.2 V      |
| 4.1 V                    | -2 dB      | 3.3 V      |
| 4.2 V                    | -3 dB      | 3.0 V      |
| 4.3 V                    | -3 dB      | 3.0 V      |
| 4.4 V                    | -3 dB      | 3.1 V      |
| 4.5 V                    | -3 dB      | 3.2 V      |
| 4.6 V                    | -3 dB      | 3.3 V      |
| 4.7 V                    | -4 dB      | 3.0 V      |
| 4.8 V                    | -4 dB      | 3.0 V      |
| 4.9 V                    | -4 dB      | 3.1 V      |
| 5.0 V                    | -4 dB      | 3.2 V      |

**Table 1: Selection of attenuators for a range of pulsed light source electrical output voltages.** Your PF32 Camera has been configured for +3.3 Volts SYNC input. A selection of voltage amplitudes  $V_{light}$  of the pulsed light source is shown in the left column, the recommended attenuator is in the second column and the actual voltage the SYNC input experiences  $V_{SYNC}$  is in the third column. Use this table to select the optimal attenuator to match specific pulsed light source to the PF32 SYNC input.



tors screw between the SMA-ended RG-316 coaxial cable coming from the pulsed light source and the SYNC input of the PF32 camera. The attenuator is shown in **Figure 4**.



**Figure 4: Mini-Circuits coaxial attenuator.** The coaxial attenuator from the FW series is used to match the electrical output of the pulsed light source to the specified SYNC input voltage. The coaxial attenuators fit between the SMA-ended RG-316 cable coming from the pulsed light source and the PF32 Camera SYNC input connector.

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## 6 Pulse Width

The PF32 Camera will not work with electrical pulses that have a pulse width of less than 3.0 ns applied to the SYNC input. If your light source produces shorter electrical pulses, contact Photon Force or your distributor. They will discuss your experimental conditions and may suggest using a pulse stretcher.



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## 7 Low Voltage at the Pulsed Light Source Electrical Output

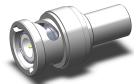
The electrical signal produced by your pulsed light source may be lower than the specified SYNC input voltage. This may stop the PF32 Camera from being able to detect the SYNC signal or increase the timing jitter. If this becomes a problem, contact Photon Force or your distributor. They will discuss your experimental conditions and may suggest using a high-bandwidth amplifier.



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## 8 Appendix A: Recommended Parts

Below are listed parts with manufacturers and part numbers, which may be needed to ensure the PF32 Camera can be safely connected to the pulsed light source.

- |                                       |                                                                                     |                                          |
|---------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------|
| • SMA-ended RG-316 coaxial cables     |    | Amphenol 135101-01<br>L-Com CCS316       |
| • 50 $\Omega$ SMA to BNC adapter      |    | Amphenol 901-165<br>L-Com BA38           |
| • 50 $\Omega$ BNC Tee adapter         |    | Amphenol 031-208-1051<br>L-Com BA250E    |
| • 50 $\Omega$ terminator              |    | Amphenol 000-46650-51RFX<br>L-Com BTB5MD |
| • 50 $\Omega$ feed-through terminator |   | Thorlabs T4119<br>Broadwave 854-009-FTT  |
| • RF attenuator                       |  | Mini-Circuits FW<br>CenticRF C18S2       |